

Scalability & Future Plan

Date	03 October 2026
Team ID	SWTID-2026-7859
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

How EduGenie can scale to more users and features, and the roadmap beyond the current version.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Depends on Gemini API availability and rate limits	Requests may fail or slow down under heavy use	High
2	No user accounts or saved history	Learners can't track progress over time	Medium
3	Runs locally (not hosted); English-only, text-only input	Limited reach; no voice/image/PDF input	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Uvicorn worker; 50 concurrent users with 0% errors (server overhead ~12-58 ms)	Containerize (Docker) and deploy to Google Cloud Run with auto-scaling and multiple workers
2	Data Storage	Stateless - no database	Add Firestore / Cloud SQL for user profiles, quiz history and progress
3	Performance	Every request calls Gemini	Cache frequent answers and learning paths, stream responses, use a paid Gemini tier
4	Security	API key in .env; no authentication	Google Sign-In / OAuth, Secret Manager for keys, rate limiting per user, HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment, user accounts, saved quiz history, progress dashboard		Anytime access and visible learning progress
Phase 3	Voice interaction, multilingual support, image/PDF input for doubt solving		Accessibility for more learners and languages
Phase 4	Mobile app, gamification (badges, streaks), teacher/parent dashboards, LMS integration (Moodle, Google Classroom)		Higher engagement and adoption by institutions